

Package #167

adopt the rapid■trigger principle from industrial gas■leak detection (continuous sensor sampling with instant alarm) to infusion pumps by adding a dual■sensor (pressure and flow) high■speed comparator that fires an alarm within milliseconds of pressure rise

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'occlusion detection delay' concerning infusion pump. User-submitted engineering problem (session input): "Why do infusion pumps fail to detect downstream occlusion in time? (provider timeout behavior)". [fda_maude] 100 records retrieved live (1997-2076). Illustrative records: 'MAUDE report 100000: Injury [VITAL SIGNS]'; 'MAUDE report 10000000: Malfunction [630G INSULIN PUMP MMT-1715K 630G BLACK MG]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2026-2026). Illustrative records: 'Telemetric lumbar cerebrospinal fluid pressure monitoring for spaceflight: a novel approach to spaceflight-associated neuro-ocular syndrome research.'; 'Modeling and Design of a Soft Capacitive Slip Sensor with Fluid Dielectric Interlayer.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: ischemic duration directly determines infarct size and clinical outcomes

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.